Sundial: A Family of Highly Capable Time Series Foundation Models

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Background: Why probabilistic time-series forecasting matters

- We see time series drive decisions in energy, finance, logistics, healthcare, and weather.
- We need calibrated uncertainty, not just point forecasts.
- We aim to generalize across domains, handle long contexts, and respond fast.

The problem: Why we need native, scalable probabilistic forecasting

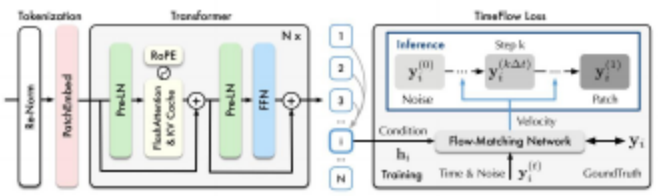
- We need native, scalable probabilistic forecasting for multimodal futures.
- We show parametric likelihoods restrict uncertainty and discrete tokenization harms context and calibration.
- We propose a native continuous, flexible, scalable generative approach.

We propose Sundial: core idea and contributions

- We introduce Sundial, a native continuous time-series foundation model that forecasts distributions.
- We propose TimeFlow Loss: conditional flow-matching for next-patch distributions without priors.
- We build an efficient decoder-only Transformer with continuous patch tokens and multi-patch prediction.
- We pre-train on a large corpus (TimeBench) and show strong zero-shot generalization.

Method overview: From continuous patches to generative forecasts

- We tokenize continuous patches and apply a causal decoder Transformer.
- We optimize TimeFlow Loss to learn next-patch distributions via flow-matching.
- At inference, we push forward from noise and sample multiple futures efficiently.



We summarize the architecture: patch tokenization + causal decoder + multi-patch prediction.

Data preparation and continuous patch tokenization

- We stationarize each instance (shift/outlier handling) and normalize per variable.
- We split series into length-P patches, pad and mask tails for arbitrary inputs.
- We embed values + masks with a shared MLP to D-dim tokens, shortening context vs. point-wise.

Backbone: Scalable, efficient decoder-only Transformer

- We use causal self-attention with Pre-LN and RoPE for stability and order awareness.
- We adopt FlashAttention for memory/throughput and KV cache for just-in-time generation.
- We predict multiple patches to reduce autoregressive steps in training and inference.

TimeFlow Loss: Intuition

- We learn conditional next-patch distributions without specifying a parametric prior.
- We flow-match velocities along paths from noise to target given context.
- We mitigate mode collapse and capture multimodal futures for calibrated uncertainty.

TimeFlow Loss: Training objective

- For token i, we condition the FM-Net on token representation h_i and time t.
- We interpolate $y_i^{(t)} = (1-t)\, y_i^{(0)} + t\, y_i$, where $t \sim \mathcal{U}[0,1]$, $y_i^{(0)} \sim \mathcal{N}(0, I)$.
- We regress the true velocity toward $y_i - y_i^{(0)}$ conditioned on h_i and t; same backbone as baselines.

TimeFlow Loss: Formal details

- We parameterize a conditional velocity field $v_\theta(y, t \mid h)$.
- We sample $t \sim \mathcal{U}[0,1]$, $y_0 \sim \mathcal{N}(0, I)$, and form $y(t) = (1-t)\, y_0 + t\, y$.
- We regress the true velocity toward $y - y_0$ conditioned on h and t.
- We avoid score estimation or diffusion SDEs; we use a direct flow-matching objective for next-patch distributions.

Sampling and complexity

- We integrate K steps with the learned velocity (push-forward) from Gaussian noise.
- We reuse the cached lookback representation and KV cache, so repeated samples are cheap.
- We scale with K and the number of predicted patches; multi-patch prediction reduces steps.

Inference: Fast probabilistic sampling

- We start from Gaussian noise and run K push-forward steps using the FM-Net and cached lookback.
- We resample with new noises to obtain many plausible futures and aggregate quantiles/median.
- We reuse the KV cache to enable millisecond zero-shot predictions and rapid resampling.

Experimental setup: Benchmarks and metrics

- We evaluate zero-shot on TSLib (point; MSE/MAE) and on FEV and GIFT-Eval (probabilistic; MASE/CRPS/WQL).

Key results: Zero-shot point forecasting on TSLib

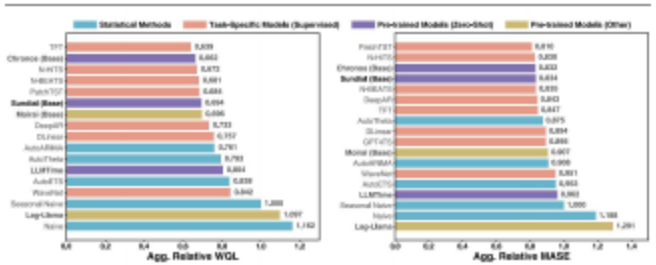
- We outperform prior SOTA Time-MoE with fewer parameters.
- We reduce avg MSE by 7.57% and MAE by 4.71% vs. Time-MoE.
- We find patch-level continuous tokens excel at long horizons.

We evaluate zero-shot on TSLib; lower MSE/MAE is better.

Models SundialSmall SundialBase SundialLarge Time-MoEBase Time-MoELarge Time-MoEUltra Timer-XL MoiraiBase MoiraiLarge ChronosBase ChronosLarge TimesFM (with respective references).
Metric MSE MAE for datasets: ETTm1, ETTm2, ETTh1, ETTh2, ECL, Weather; plus 1st Count.
ETTm1: 0.354 0.388 — 0.336 0.377 — 0.331 0.369 — 0.394 0.415 — 0.376 0.405 — 0.356 0.391 — 0.373 0.392 — 0.406 0.385 — 0.422 0.391 — 0.645 0.500 — 0.555 0.465 — 0.433 0.418
ETTm2: 0.265 0.324 — 0.258 0.320 — 0.254 0.315 — 0.317 0.365 — 0.316 0.361 — 0.288 0.344 — 0.273 0.336 — 0.311 0.337 — 0.329 0.343 — 0.310 0.350 — 0.295 0.338 — 0.328 0.346
ETTh1: 0.390 0.418 — 0.411 0.434 — 0.395 0.420 — 0.400 0.424 — 0.394 0.419 — 0.412 0.426 — 0.404 0.417 — 0.417 0.419 — 0.480 0.439 — 0.591 0.468 — 0.588 0.466 — 0.473 0.443
ETTh2: 0.340 0.387 — 0.333 0.387 — 0.334 0.387 — 0.366 0.404 — 0.405 0.415 — 0.371 0.399 — 0.347 0.388 — 0.362 0.382 — 0.367 0.377 — 0.405 0.410 — 0.455 0.427 — 0.392 0.406
ECL: 0.169 0.265 — 0.169 0.265 — 0.166 0.262 — — — — — — Other baselines: 0.174

Zero-shot probabilistic forecasting on FEV

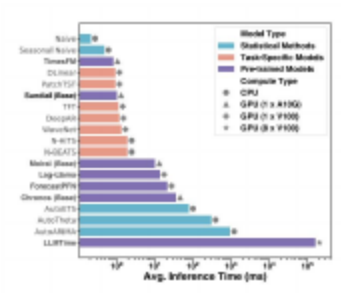
- We outperform most supervised/statistical baselines zero-shot.
- We are competitive among pre-trained models with strong MASE/WQL.
- We generate 20 samples to deliver calibrated intervals.



We evaluate zero-shot on FEV; lower MASE/WQL is better.

Inference speed: Milliseconds with just-in-time sampling

- We approach N-BEATS latency and are 35× faster than Chronos on FEV.
- We reduce steps via patch-wise and multi-patch generation.
- We cut memory/time with FlashAttention and KV cache.



We report average latency on FEV (log-scale x-axis) with annotated compute budgets.

Ablation: Training objectives (probabilistic quality)

- We achieve the best CRPS on GIFT-Eval and across datasets.
- We also improve MSE vs. diffusion and MSE objectives (see Slide "TSLib results").
- We confirm the advantage of generative distribution learning and calibration.

Table: Table 7. Zero-shot probabilistic forecasting with different training objectives. Averaged CRPS (lower is better).

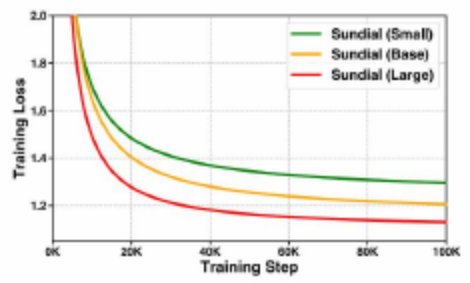
Objective	ETTh1↓	ETTh2↓	ETTm1↓	ETTm2↓	ECL↓	Weather↓	GIFT-Eval↓
TimeFlow Loss	0.0059	0.0037	0.0057	0.0029	0.0082	0.0021	0.5050
Diffusion Loss	0.0082	0.0053	0.0070	0.0039	0.0095	0.0032	0.5340
MSE Loss	0.0063	0.0040	0.0058	0.0032	0.0080	0.0023	0.6420

Engineering ablations: What makes it fast and stable?

- RoPE improves zero-shot; Pre-LN stabilizes convergence.
- Cut memory by 14.8%.
- Reduce inference time by 43.6%.

Scaling behavior: Larger models, lower loss

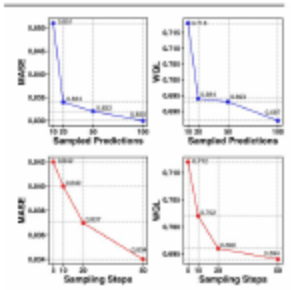
- We observe consistent capacity utilization in training curves.
- Large reduces converged loss by 15.38%.
- We benefit from scaling both data (TimeBench) and model size.



We plot TimeBench pre-training loss for different model sizes.

Test-time calibration: Accuracy–latency control

- We improve MASE and WQL by increasing samples or steps K without retraining.
- 1 second on CPU with 20 samples $\times$ 50 steps.
- We offer a flexible trade-off across deployment budgets.



We show FEV MASE and WQL trends as we vary sample count and sampling steps.

Limitations and considerations

- We currently pre-train on univariate series; multivariate/covariates are future work.
- We still find very high-frequency and ultra-long contexts challenging.
- We can further improve calibration via sampling design and post-processing.

Conclusion: What we deliver with Sundial

- We provide native, flexible generative forecasting for continuous time series.
- We avoid priors and mitigate mode collapse with TimeFlow Loss.
- We achieve state-of-the-art zero-shot accuracy with millisecond inference.
- We leverage TimeBench to study scaling laws and robust generalization.

Baselines and sources

- We compare to: Time-MoE [Shi et al., 2024b], Timer-XL [Wen et al., 2024], Moirai [Jain et al., 2024], Chronos [Rasul et al., 2024], TimesFM [Das et al., 2024], N-BEATS [Oreshkin et al., 2020].
- We evaluate on datasets: TSLib [Wu et al., 2022], FEV [Ansari et al., 2024], GIFT-Eval [Shi et al., 2024a].
- We follow author–year strings consistent with the paper's references.

Questions & discussion

- We appreciate your attention and feedback!
- We welcome questions and discussion.